

Cambridge Lower Secondary Checkpoint

CANDIDATE
NAME

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CENTRE
NUMBER

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CANDIDATE
NUMBER

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MATHEMATICS

0862/01

Paper 1

April 2024

1 hour

You must answer on the question paper.

You will need: Geometrical instruments
 Tracing paper (optional)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You should show all your working in the booklet.
- You are **not** allowed to use a calculator.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **12** pages.

- 1 Youssef thinks of a number, n .
He adds 3
His answer is greater than or equal to -5 and less than 17

Write the correct inequality signs to complete the inequality.

$$-5 \text{ } n + 3 \text{ } 17 \quad [1]$$

2 $200^{-3} = \frac{1}{200^w}$

Write the value of w .

$$w = \text{.....} \quad [1]$$

- 3 Solve the simultaneous equations.

$$\begin{aligned} x - 2y &= 2 \\ 5x + 2y &= 58 \end{aligned}$$

$$\begin{aligned} x &= \text{.....} \\ y &= \text{.....} \end{aligned} \quad [2]$$

- 4 Find the exterior angle of a regular 10-sided polygon.

$$\text{.....}^\circ \quad [1]$$

5 Here is a sequence of calculations.

$$1 \times 7 - 2 \times 1 = 5$$

$$3 \times 8 - 4 \times 2 = 16$$

$$5 \times 9 - 6 \times 3 = 27$$

$$7 \times 10 - 8 \times 4 = 38$$

$$9 \times 11 - 10 \times 5 = 49$$

Complete the next calculation in this sequence.

$$11 \times 12 - \dots \times \dots = \dots$$

[1]

6 A regular polygon has k lines of symmetry.

Tick (✓) the correct statement about the order of rotational symmetry of the polygon.

The order of rotational symmetry is 1

☐

The order of rotational symmetry is clockwise

☐

The order of rotational symmetry is k

☐

The order of rotational symmetry is $k + 1$

☐

[1]

- 7 (a) Mike and Pierre are each asked to write the equations of two lines that have a **positive** y -intercept.

Tick (✓) to show if each student is correct or not correct.

		Both equations have a positive y -intercept	
		Correct	Not correct
Mike			
$y = -x + 3$	$y = 2 - x$	☐	☐
Pierre			
$y = x + 0.5$	$y = 7 - 5x$	☐	☐

[1]

- (b) The equation of a line is $7 = 3x + y$

Find the gradient and the y -intercept of this line.

gradient =

y -intercept =

[2]

- 8 Tick (✓) to show if each conversion is correct or not correct.

	Correct	Not correct
60 nm = 6 mm	☐	☐
2000 GB = 2 MB	☐	☐

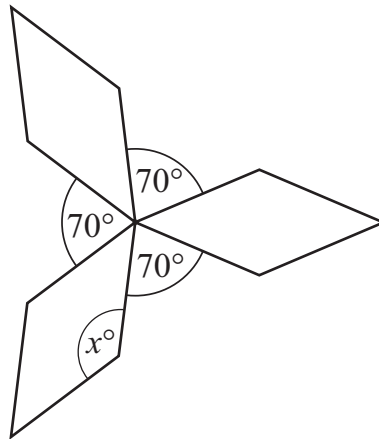
[1]

- 9 Represent $-2 < x \leq 4$ on the number line.



[1]

- 10 The diagram shows a shape made with three identical rhombuses.



NOT TO
SCALE

Find the value of x .

$x =$ [4]

- 11 Gabriella collects the heights of 11 indoor plants and 12 outdoor plants. The table and the incomplete back-to-back stem-and-leaf diagram show information about her results.

Outdoor plants	
Modal height	39 cm
Minimum height	17 cm
Range	26 cm

Indoor plants							Outdoor plants					
			9	8	7	0						
		9	9	5	4	1						
			9	8	6	2	3	7	7			
					0	3	4	5	8	9		
						4	1					

Key: 6 | 2 | 3 represents indoor plant height of 26 cm
and outdoor plant height of 23 cm

- (a) Use the information in the table to complete the back-to-back stem-and-leaf diagram.

[3]

- (b) Calculate the range of heights for the indoor plants.

..... cm [1]

- 12 Work out the value of $\frac{n^4 + 29}{n + 7}$ when $n = 3$

..... [2]

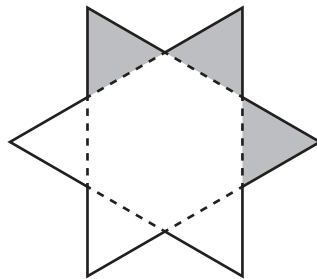
13 (a) Write 62 000 in standard form.

..... [1]

(b) Write 8.1×10^{-3} as an ordinary number.

..... [1]

14 The diagram shows a shape made from 6 congruent equilateral triangles and a regular hexagon.



The regular hexagon has an area of 140 cm^2 .

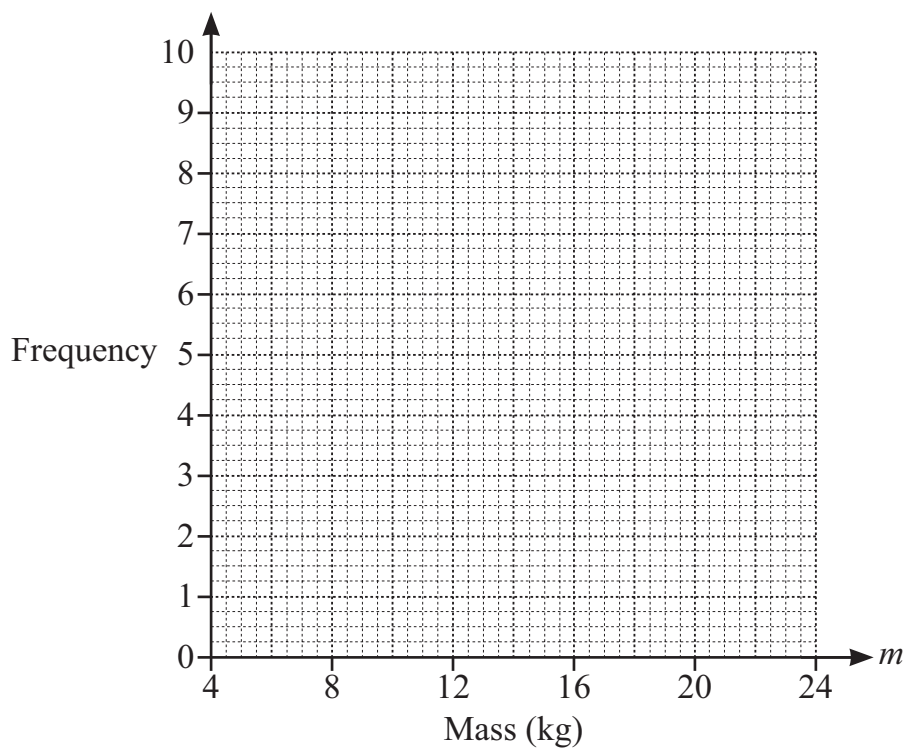
Calculate the shaded area.

..... cm^2 [1]

15 The table shows information about the mass of each of 23 boxes.

Mass (m , kg)	$4 \leq m < 8$	$8 \leq m < 12$	$12 \leq m < 16$	$16 \leq m < 20$	$20 \leq m < 24$
Frequency	8	7	3	2	3

(a) On the grid, draw a frequency polygon to show this information.



[3]

(b) Draw a ring around the interval that contains the median mass.

$4 \leq m < 8$ $8 \leq m < 12$ $12 \leq m < 16$ $16 \leq m < 20$ $20 \leq m < 24$

[1]

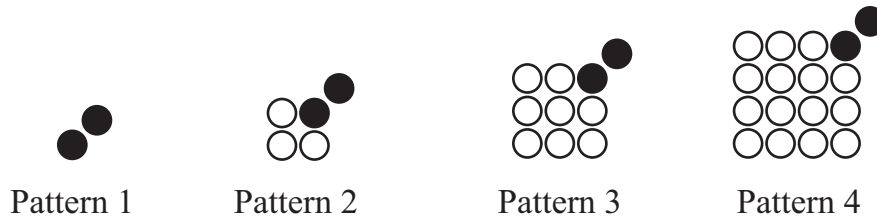
16 $\frac{1}{9} = 0.\dot{1}$

Use this fact to convert $\frac{7}{9}$ to a decimal.

Give your answer correct to 3 decimal places.

..... [2]

17 Jamila makes this sequence of patterns using white counters and black counters.



(a) Complete these sentences.

The first one has been done for you.

The number of white counters in pattern 4 is 15

The number of white counters in pattern 5 is

The number of white counters in pattern 100 is

[2]

(b) Write an expression, in terms of n , for the **total** number of counters in pattern n .

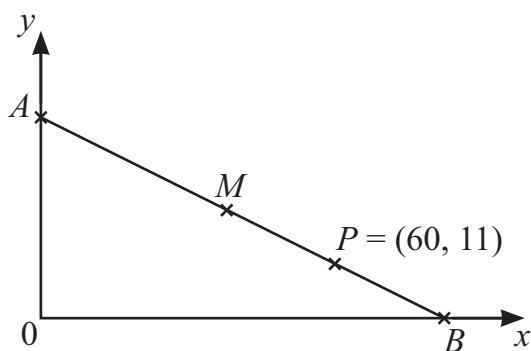
..... [2]

18 Tick (✓) to show if each of the calculations is equivalent to 37×10^{-3} or not.

Calculation	Equivalent to 37×10^{-3}	Not equivalent to 37×10^{-3}
3.7×0.01	☐	☐
3.7×10^{-2}	☐	☐
3.7×10^{-4}	☐	☐
$37 \div 10^3$	☐	☐

[2]

19 Point A lies on the y -axis and point B lies on the x -axis.



NOT TO
SCALE

M is the midpoint of AB .

P is the midpoint of MB .

Find the coordinates of point A and the coordinates of point B .

$A = (\dots\dots\dots, \dots\dots\dots)$

$B = (\dots\dots\dots, \dots\dots\dots)$

[2]

- 20** x is an integer and $1 < \sqrt[3]{x} < 2$

Complete these sentences about x .

One possible value of x is

There is a total of possible values of x .

[2]

- 21** Point P lies on the line $y = 5 - 2x$

The x -coordinate of P is a negative integer.

The y -coordinate of P is a prime number.

Find a possible pair of coordinates for point P .

$P = (\dots\dots\dots, \dots\dots\dots)$ [2]

22 Work out.

$$\frac{8}{9} + 1\frac{1}{3} \div 4\frac{4}{5}$$

Give your answer as a mixed number in its simplest form.

..... [4]

23 A square has a side length of x cm.

Mia enlarges this square by increasing each side length of the square by 200%.

(a) The side length of her enlarged square measures kx cm.

Draw a ring around the value of k .

2 3 4 200 300

[1]

(b) Find the percentage increase in the area of the square after enlargement.

..... % [2]

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